TNTIC 2026 TAMIL NADU TOURISM INNOVATION CHALLENGE Stage: Working Prototype (POC)

SMART AGRO-TOURISM & RURAL HERITAGE INTELLIGENCE

THINAI திணை

Empowering Tamil Nadu's Rural Agro-Tourism Circuits with Offline-First AI Crop Intelligence & Cultural Landscape Storytelling.

Sangam 5-Landscape Heritage | Offline AI Computer Vision | ARFI Resilience Modeling | Farm-to-Fork Traceability

Innovator: Swastikan S swastikan.2401246@srec.ac.in

The Rise of Experiential Rural & Agro-Tourism

Post-pandemic travelers seek authentic, nature-immersive farm stays, eco-trails, and agrarian culinary heritage rather than crowded destinations.

32% YoY

Growth in Eco & Farm-Stays

Travelers flocking to Pollachi, Nilgiris, Valparai, Kolli Hills, and Thanjavur delta for authentic organic farm experiences.

7 Agro-Zones

Tamil Nadu's Natural Asset

From spice plantations to delta paddy fields, Tamil Nadu possesses the richest historical agricultural tapestry in South India.

2.4x Revenue

Farmer Income Diversification

Integrating experiential farm tourism gives smallholder farmers steady secondary hospitality revenue against crop failure.

Why Rural Agro-Tourism Struggles Today

Rural farms lack the digital tools to offer modern, informative tourist experiences while maintaining crop health under shifting weather.

1

Dead Zones Break Cloud Apps

Rural farm-stays and plantation trails experience frequent cellular dead zones. Standard cloud-based tourism and agri apps fail completely without internet.

2

No Interactive Interpretation

Tourists walk through farms blindly without understanding crop lifecycles, organic techniques, medicinal flora, or ancient Tamil farming heritage.

3

Crop Fragility Risks

Sudden pest outbreaks and microclimate stress ruin demonstration fields and orchards, destroying the aesthetic and financial foundation of the farm stay.

THINAI: The Intelligent Agro-Tourism Operating System

A dual-purpose platform built for both the **eco-traveler** and the **farm-stay host**, operating seamlessly 100% offline.

FOR TOURISTS & TRAVELERS — Immersive Experience

The Digital Farm Companion

- ✓ **Point-and-Scan Crop Doctor:** Instant offline AI leaf/crop identification and botanical insights right from smartphone camera.
- ✓ **Sangam Eco-Trails (Aintinai):** Interactive cultural storytelling exploring Kurinji, Mullai, Marutham, Neithal, and Palai landscapes.
- ✓ **Farm-to-Fork Mandi Transparency:** Verified local price benchmarks to buy authentic organic produce directly from farmers.

FOR RURAL FARM-STAY HOSTS — Operational Resilience

Smart Farm Health & Decision Engine

- ✓ **ARFI Stress-Testing:** Predicts recommendation fragility under sudden rainfall or heatwaves to preserve show crops.
- ✓ **100% Offline-First Architecture:** Dexie IndexedDB + Background Sync ensures continuous operation in remote hills and forests.
- ✓ **8-Stage Lifecycle Guide & Fertilizer Calculator:** Scientific organic nutrient schedules calibrated to farm soil tests.

Engineered for Remote Resilience & Real Intelligence

Built without placeholder code. Transparent decision pipelines running locally on modern web & native Android mobile devices.

01 / VISION ENGINE

Offline Crop Doctor

PlantVillage CV pathology running on-device. Evaluates leaf boundary sharpness, lesion discoloration, and pathogen confidence in 250ms.

02 / ALGORITHMIC CORE

ARFI Fragility Index

Evaluates how robust a farm action is under shifting microclimates (+/- 20% rainfall, humidity spikes), offering safe alternative actions.

03 / DATA LAYER

Offline-First Dexie DB

Browser IndexedDB + mutation queues. Automatic sync when internet returns with transparent LIVE / CACHED / OFFLINE data provenance badges.

04 / NATIVE HYBRID

Capacitor Android

Compiled native APK (`thinai-debug.apk`) with rear camera hardware integration, multi-language support (Tamil/English), and fast Vite runtime.

Transforming Tamil Nadu into a Global Agro-Tourism Destination

Aligning state tourism goals with grassroots rural empowerment and environmental preservation.

A

Expanding Tourist Circuits

Decongests traditional heritage centers by routing urban & international tourists into scenic, sustainable rural agro-circuits across Kongu, Delta, and Western Ghats.

B

Grassroots Economic Empowerment

Provides smallholders and village communities with sustainable hospitality income, farm-gate organic sales, and cultural artisan opportunities.

C

Preserving Agrarian Heritage

Digitizes indigenous seeds, heritage grains (Mappillai Samba, Karuppu Kavuni), and organic cultivation methods for global educational appreciation.

Tested, Functional & Demonstrable Today

THINAI is not a concept wireframe—it is an operational system tested on live microclimate and crop datasets.

THINAI Core Engine

Tri-pillar risk synthesis: Water, Disease, and Market risk with causal reasons.

Camera Crop Doctor

Rear camera hardware capture + leaf contrast validation + offline pathology.

ARFI Stress Lab

Simulate drought, flood, and price crash to test decision resilience.

Offline Sync Bar

Toggle offline mode; auto-queues transactions in IndexedDB for auto-sync.

Native Android APK Build: `thinai-debug.apk` (Capacitor 6.x / Android 14 Ready)

Production Web Build: **✓ 2,664 Modules Transformed (0 Errors)**

Sustainable Revenue & Community Value Creation

A multi-stakeholder model designed for low-cost rural adoption and high-value tourist experiences.

B2B SAAS

Farm-Stay Subscriptions

Tiered subscription for agro-tourism hosts: includes custom digital trail QR codes, crop diagnostic portal, and automated guest digital guides.

MARKETPLACE

Farm-to-Tourist Produce

Zero-commission for smallholders with small nominal convenience fee on premium certified organic produce shipped to urban tourist homes.

INSTITUTIONAL

TTDC & Eco-Circuit Licences

Partnership integration with Tamil Nadu Tourism Development Corporation (TTDC) to power official state agro-tourism trails and visitor interpretation kiosks.

09. EXECUTION ROADMAP

2026 - 2027 Milestones

From Campus Prototype to Statewide Agro-Trails

Structured phased rollout starting with the Kongu agro-tourism belt.

PHASE 1 (Q4 2026)

Prototype & Field Testing

Deploy offline APK across 10 rural farm stays in Pollachi, Coimbatore, and Nilgiris foothills with live tourist feedback.

PHASE 2 (Q1 2027)

Agro-Trail QR Ingestion

Onboard 50 certified organic farm stays; introduce dynamic trail guides and multi-dialect Tamil/English audio guides.

PHASE 3 (Q2 2027)

TTDC Integration

Link with Tamil Nadu Tourism portal; launch curated Sangam Aintinai circuits for inbound domestic & international tourists.

PHASE 4 (Q3 2027)

Pan-State Expansion

Scale to 250+ rural agrarian partners across Thanjavur Delta, Madurai, and Southern coastal circuits.

Empowering Tamil Nadu's Land, People & Heritage

We are building THINAI to prove that technological innovation does not replace agrarian tradition—it protects, celebrates, and sustains it.

SS

Swastikan S

Student Innovator • Engineering Scholar

Sri Ramakrishna Engineering College (SREC), Coimbatore

Email: swastikan.2401246@srec.ac.in

Focus: Offline-First Mobile Architectures, Applied Computer Vision & Agronomic Decision Modeling

